Dual-Device (Personal + Corporate) Management with Single Sign-On in Microsoft Intune

Implementation Solution Guide

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Executive Summary — Dual-Device (Personal + Corporate) Management with Single Sign-On

A huge share of your people carry **two devices** — a **corporate-owned device (COD)** the company buys and controls, and their **own personal device (BYOD)** they also use for work. They expect the **same work account to just sign in on both**, seamlessly. Handled badly, that means duplicate policies, inconsistent security, two logins, and frustrated users.

This solution is built for exactly that **one-person-two-devices** scenario, under **Microsoft Intune** and **one Microsoft Entra identity**: the user signs in with the same corporate account on **both** their company device and their personal device, gets **single sign-on** across them, and IT applies the **right level of control to each** based on who owns it.

- **Corporate-owned (COD)** → full MDM enrollment, device compliance, and configuration. IT owns the device end to end.
- **Personal / BYOD** → lighter-touch management: **App Protection (MAM)** that secures corporate data inside apps without taking over the employee's device — or optional personal MDM enrollment.
- **One identity, both devices, single sign-on** → the user's single Entra account works on both of their devices; the Microsoft identity broker (WAM on the corporate device, Microsoft Authenticator on the personal device) gives **silent SSO** so they sign in once per device. **Ownership-based targeting** routes the correct policy to each device automatically, and **Conditional Access** grants access from their compliant company device *or* a protected app on their personal one.

Built the standard repeatable way — **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** — with exact, click-by-click Intune steps, a requirements guide that tells you how to verify *and* fix each prerequisite, five ready-to-run PowerShell scripts (Microsoft Graph), and testing/troubleshooting guides with clear pass/fail gates.

Outcome: one employee account that signs in seamlessly — with single sign-on — across both their corporate and personal device, corporate data protected on each, and access governed by compliance and app protection — delivered by your own team without a consulting engagement.

1. Solution Overview — Dual-Device (Personal + Corporate) Management with Single Sign-On

What this solution is

This solution targets a specific, very common reality: **one person carries two devices** — a **personal device (BYOD)** they own and a **corporate-owned device (COD)** the company owns — and needs the **same single work account to sign in seamlessly on both**, with the right level of management applied to each. Think of the executive with a corporate laptop and a personal phone, the field tech with a company tablet and their own handset, or the hybrid employee who answers Teams on their personal device and does focused work on the corporate one.

It delivers that in Microsoft Intune under **one Microsoft Entra identity**, across the two ownership models:

1. **Corporate-Owned Device (COD)** — the device the organization buys and owns. Fully enrolled in Intune (MDM), receives configuration and compliance policies, managed end to end by IT.

1. **Personal Device (BYOD)** — the device the employee owns. The corporate **identity and data** are protected without taking over the personal device, primarily through **App Protection Policies (MAM)**, with optional lightweight personal MDM enrollment.

The defining design goal: **the user signs in with one Entra account on both of their devices and gets a single sign-on experience** — Intune applies controls based on **device ownership**, not on a second account, so the personal device is protected (data-only) while the corporate device is fully managed, simultaneously, for the same person.

How it works

1. **One identity, two devices.** The user has a single Microsoft Entra account. That one account registers/enrolls their corporate device **and** signs into the managed work apps on their personal device — there is no separate "BYOD account," and both devices register to the same identity.

1. **Single sign-on across both.** Once the account is established on a device, the **Microsoft identity broker **(Microsoft Authenticator / Company Portal on the personal device; Entra-joined WAM on the corporate device) gives the user** silent SSO** into Microsoft 365 and SSO-enabled apps on each device — sign in once per device, then move between them without re-entering credentials.

1. **Ownership is tagged.** Every device is marked **Corporate** or **Personal** (by enrollment method, corporate device identifiers, or enrollment-restriction defaults).

1. **Ownership-based targeting.** Policies are assigned to the **user**, then scoped with **device filters** on `deviceOwnership` so the user's two devices each get the correct set:

• the user's **Corporate** device receives full **device compliance** + **configuration** policies;

• the user's **Personal** device receives **App Protection (MAM)** policies that encrypt corporate data, enforce a PIN, and block copy/paste or "Save As" to personal locations.

1. **Conditional Access ties it together.** A single Conditional Access policy for the user grants access when the request comes from their **compliant corporate device** *or* from an **approved app with an app-protection policy** on their personal device — so the one account works securely from **either** of the user's two devices.

Single sign-on across the user's two devices

Because both devices register to the **same Entra identity**, the user gets a unified sign-in:

• **Corporate device:** Entra-joined (or hybrid-joined); Windows sign-in / the WAM broker provides primary-refresh-token (PRT) based SSO to Microsoft 365 and Entra apps — no repeated passwords.

• **Personal device:** the **Microsoft Authenticator** broker (registered via Company Portal during app-protection onboarding) brokers Entra auth, so Outlook/Edge/Teams and other approved apps get silent SSO with the same account.

• **Consistent posture either way:** Conditional Access evaluates each device on its own merits

(compliant COD *or* MAM-protected BYOD), so the single account is honored from both without the user ever managing two logins.

Management models (choose per ownership)

• **COD — full MDM:** complete device management, configuration profiles, compliance, and (optionally)

Autopilot/ADE/Android Enterprise corporate enrollment for zero-touch provisioning.

• **BYOD — App Protection / MAM (recommended):** secures corporate data *inside* managed apps with no

full device enrollment; least intrusive for employees, strong data-leak protection.

• **BYOD — personal MDM enrollment (optional):** when a personal device must receive a few device

settings; still tagged Personal and scoped accordingly.

This solution standardizes on **full MDM for COD** and **App Protection for BYOD**, with personal MDM called out as an option.

Benefits

• **One account on both of the user's devices** — no duplicate identities; the same login on the

personal and the corporate device.

• **Seamless single sign-on** — sign in once per device, then move between Microsoft 365 apps (and

between the two devices) without re-entering credentials.

• **Right-sized control** — full management on the corporate device; data-only protection on the

personal one — at the same time, for the same person.

• **Lower friction BYOD** — employees keep their personal device private; corporate data stays

contained in managed apps.

• **Consistent security** — compliance for COD, app protection for BYOD, unified by one Conditional

Access policy that accepts either device.

• **Scales cleanly** — ownership-based targeting means you assign once to the user and the right policy

follows each of their devices.

Limitations (be honest with buyers)

• **App Protection (MAM)** secures data **inside supported apps** (Microsoft 365 and SDK-enabled

apps); it does not manage the whole personal device or unmanaged apps.

• BYOD users must use the **managed/approved apps** (e.g. Outlook, Edge, Teams) to get corporate data;

built-in personal mail/browser apps are blocked from corporate data by design.

• COD zero-touch provisioning (Autopilot/ADE/Android Enterprise) is referenced but each platform's full enrollment is covered by its own solution; this solution unifies the **model**.

• A user signing in on a brand-new device enrolls/registers that device the first time.

The standard deployment model

This solution uses **one** model so the process is repeatable across our catalog:

> ***One Microsoft Entra identity per user, signed in across that user's two devices with single sign-on; the corporate-owned device under full MDM and the personal device under App Protection (MAM); ownership-based targeting routes the right policy to each; a single Conditional Access policy governs access from both.***

Every step follows the repeatable pattern: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

2. Requirements — Dual-Device (Personal + Corporate) Management with Single Sign-On

Confirm **every** requirement below before you begin. Each entry follows the same pattern: **Description · Why required · How to verify · How to implement if missing.** Category-wide prerequisites (tenant, Intune licensing, admin roles, pilot group, supported OS, network) are covered once in the **Global Requirements — Intune Solutions** document; the items here are specific to running COD and BYOD under one identity.

R1. A single Microsoft Entra identity per user

Description: Each user has exactly one Entra ID account used for both device types. **Why:** The whole design depends on one account signing in on both the corporate and the personal device; two accounts would split policy and defeat the goal. **Verify:** Entra → **Users**; the pilot user exists once with a licensed account. **Implement:** Consolidate duplicate accounts; ensure the user signs in with the same UPN everywhere.

R2. Intune licensing that includes App Protection (MAM)

Description: A license plan that covers both device management (MDM) and app management (MAM). **Why:** COD uses MDM; BYOD uses App Protection (MAM). Both must be licensed for the same user. **Verify:** Entra → user → **Licenses**; an Intune-bearing plan (M365 E3/E5, Business Premium, or Intune Plan 1) is assigned. **Implement:** Assign the appropriate Microsoft 365/Intune license to all pilot users.

R3. Microsoft Entra ID P1 (for Conditional Access)

Description: Entra ID P1 (or higher) for Conditional Access. **Why:** A single Conditional Access policy is what lets one account access work from a compliant COD *or* an app-protected BYOD device; CA requires P1. **Verify:** Entra → **Conditional Access** opens and policies can be created (P1 present). **Implement:** Assign Entra ID P1 (included in M365 E3/E5/Business Premium) to pilot users.

R4. Enrollment restrictions configured for both ownership types

Description: Device platform/enrollment restrictions that **allow personal (BYOD)** enrollment for the BYOD path and treat the correct platforms as **corporate** for COD. **Why:** A restriction that blocks personal enrollment, or that auto-marks all devices corporate, breaks the ownership distinction this solution relies on. **Verify:** Intune → **Devices** → **Enrollment** → **Enrollment device platform restrictions**; the pilot restriction shows the intended **Personal = Allow/Block** per platform. **Implement:** Edit the default or a pilot-targeted restriction so BYOD platforms allow personal, and keep COD platforms enrolling as corporate (see R5).

R5. A way to mark devices Corporate

Description: Corporate device identifiers and/or corporate enrollment methods so COD devices are tagged **Corporate**. **Why:** Ownership = Corporate is the signal that routes full MDM policy (vs. App Protection) to a device. **Verify:** Intune → **Devices** → **Enrollment** → **Corporate device identifiers** (serials/IMEIs), or a device's **Ownership** field reads **Corporate**. **Implement:** Add corporate identifiers, use corporate enrollment programs (Autopilot/ADE/Android Enterprise corporate), or set **Ownership = Corporate** on the device record.

R6. Company Portal app

Description: The Intune **Company Portal** available on managed devices. **Why:** Company Portal performs MDM enrollment for COD and is the broker that registers a personal device and surfaces app-protection-enabled apps for BYOD. **Verify:** Intune → **Apps**; Company Portal is added for the relevant platforms and assigned to the pilot group. **Implement:** Add Company Portal per platform and assign it to the pilot group.

R7. App Protection (MAM) capability and approved apps

Description: App Protection Policies plus the approved managed apps (Outlook, Edge, Teams, Office). **Why:** BYOD data protection lives in App Protection Policies applied to managed apps; without the apps, BYOD users can't safely reach corporate data. **Verify:** Intune → **Apps** → **App protection policies** can be created; the approved apps are deployed or available. **Implement:** Create App Protection Policies (Implementation §4) and make the approved apps available to BYOD users via Company Portal / store.

R8. Device compliance policy for corporate devices

Description: A compliance policy targeting corporate-owned devices. **Why:** Conditional Access for COD grants access only to **compliant** devices; the policy defines compliant. **Verify:** Intune → **Devices** → **Compliance**; a policy exists and is assigned (scoped to Corporate). **Implement:** Create a compliance policy (OS minimums, encryption, etc.) and target it with the Corporate device filter (Implementation §5).

R9. Device-ownership filters

Description: Intune **device filters** on `deviceOwnership` (Corporate / Personal). **Why:** Filters are how one assignment to a user group applies the *right* policy to the *right* device type — the mechanism that makes "one account, both devices" work cleanly. **Verify:** Intune → **Tenant administration** → **Filters**; filters for Corporate and Personal exist. **Implement:** Create them in Implementation §5, or run the included `Set-OwnershipBasedTargeting.ps1`.

R10. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, app protection check-ins, and authentication depend on these. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level on the marketplace so individual solution listings stay small. Confirm these are in place before deploying *any* Intune solution.

Each entry: **Description · Why required · How to verify · How to implement if missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, enrollment, and security policies. **Verify:** `https://intune.microsoft.com` loads and **Tenant administration** → **Tenant status** shows an active subscription. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies, enrollment connectors, and configuration profiles requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator** (or Global Administrator for connector setup). **Implement:** Assign **Intune Administrator** to the operator; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted devices are enrolled and manageable in Intune. **Why:** Policies only reach enrolled (or app-managed) devices, regardless of platform. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Configure the appropriate enrollment method for the platform and ownership model, then enroll the device.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted to a small group before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists with intended members. **Implement:** Entra → **Groups** → **New group** → type **Security** → add pilot members.

G6. Supported, current operating system

Description: A supported OS baseline with current updates for the targeted platform. **Why:** Modern Intune features and security settings assume a patched, vendor-supported OS. **Verify:** Check the device's OS version or Intune → device → **Hardware**. **Implement:** Deploy update policies or upgrade to a supported release.

G7. Network access to required cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation — Dual-Device (Personal + Corporate) Management with Single Sign-On

Follow these steps **in order**. The *why* lives in the Solution Overview; the steps below are short and exact so you can **click to implement**. The whole design assumes **one Entra identity per user** that works on both device types — so you assign policies to the **user (group)** and let **ownership-based filters** route the right policy to each device. Pilot first: **Assess → Configure → Pilot → Validate → Expand.**

All steps use the Microsoft Intune admin center at `https://intune.microsoft.com` and Microsoft Entra at `https://entra.microsoft.com` unless noted.

Phase 1 — Identity & enrollment baseline

Step 1. Confirm one identity and licensing per pilot user

1. Entra → **Users** → confirm each pilot user is a **single** account with an Intune + Entra ID P1

license (Requirements R1–R3).

1. Entra → **Groups** → **New group** → **Security** → name it `PILOT – COD-BYOD Users` → add the pilot users.

*(You assign every policy in this solution to this **user** group.)*

Step 2. Set enrollment restrictions for both ownership types

1. **Devices → Enrollment → Enrollment device platform restrictions.**
2. Open the default (or create a pilot-scoped restriction **named** `Enrollment Restriction – COD-BYOD – Pilot`).
3. For **BYOD** platforms (e.g. iOS/Android personal), set **Personal devices = Allow**.
4. For platforms you only allow as corporate, set **Personal devices = Block** (those must enroll via

a corporate method).

Step 3. Establish how COD devices are marked Corporate

1. **Devices** → **Enrollment** → **Corporate device identifiers** → **Add** serial numbers/IMEIs of company devices **(or)** rely on a corporate enrollment program (Autopilot/ADE/Android Enterprise).
1. Confirm a known company device shows **Ownership = Corporate** on its device record.

Phase 2 — Ownership-based targeting (the core of "one account, both devices")

Step 4. Create device-ownership filters

1. **Tenant administration** → **Filters** → **Create.**
2. Filter 1: **Corporate Devices** — platform as needed; rule `(device.deviceOwnership -eq "Corporate")`.
3. Filter 2: **Personal Devices** — rule `(device.deviceOwnership -eq "Personal")`.
4. **Save** both.

> *Prefer automation? Run `Set-OwnershipBasedTargeting.ps1` to create these filters (and optionally assign the policies below) via Microsoft Graph.*

Phase 3 — Corporate-owned (COD): full MDM

Step 5. Create a compliance policy for corporate devices

1. **Devices** → **Compliance** → **Create policy** → pick the platform; **name it** `COD – Compliance – Pilot`.
2. Set baseline rules (OS minimum, disk encryption, firewall, etc.).
3. **Assignments:** assign to **PILOT – COD-BYOD Users**, and under **Filter**, **Include** the **Corporate Devices** filter → **Create**.

Step 6. Create configuration profile(s) for corporate devices

1. **Devices** → **Configuration** → **Create** → **New policy** → platform → your settings; **name it** `COD – Configuration – Pilot`.
2. **Assignments:** assign to **PILOT – COD-BYOD Users**, **Include** the **Corporate Devices** filter.

(Zero-touch corporate enrollment — Windows Autopilot, Apple ADE, Android Enterprise corporate — is each covered by its own solution; plug those in for fully automated COD provisioning.)

Phase 4 — BYOD: App Protection (MAM)

Step 7. Create App Protection Policies

1. **Apps** → **App protection policies** → **Create policy** → **iOS/iPadOS**, **name it** `BYOD – App Protection – iOS – Pilot` (repeat for **Android** as `BYOD – App Protection – Android – Pilot`).
2. **Apps:** select the approved managed apps (Outlook, Edge, Teams, Office, OneDrive).
3. **Data protection:** block **backup**, restrict **cut/copy/paste** to managed apps, block **Save As** to personal locations, require corporate data to stay in managed apps.
1. **Access requirements:** require **PIN** / org credentials to access; set offline timeouts.
2. **Assignments:** assign to **PILOT – COD-BYOD Users**. *(App Protection is identity-based — it applies to the user wherever they sign into the managed app, so it naturally targets the personal device without device enrollment.)*

Step 8. Make the approved apps available to BYOD users

1. **Apps** → **(platform)** → **Add** Outlook, Edge, Teams, Office, Company Portal as needed.
2. Assign them **Available** to **PILOT – COD-BYOD Users** so users install them from Company Portal / store.

Phase 5 — Unify access with Conditional Access

Step 9. One Conditional Access policy for both device types

1. Entra → **Protection** → **Conditional Access** → **Create new policy** → name it `CA – COD-BYOD Access`.
2. **Users: PILOT – COD-BYOD Users**.
3. **Target resources:** Office 365 (and other corporate apps).
4. **Grant:** require **one of the following** — **Require device to be marked as compliant** *(this is satisfied by the corporate device)* **and/or Require app protection policy** *(satisfied by the personal device's managed app)*; set the control to **Require one of the selected controls**.
1. **Enable policy: Report-only** for the pilot → **Create**.

This single policy is what lets the **same account** reach corporate data from a **compliant COD** or an **app-protected BYOD** device.

Phase 5b — Single sign-on across the user's two devices

The goal here is that the user authenticates **once per device** and then gets **silent SSO** to Microsoft 365 and Entra apps on **both** of their devices, with the same account.

Step 9a. Corporate device — broker-based SSO

1. Ensure the corporate device is **Microsoft Entra joined** (or Hybrid Entra joined) during enrollment — this gives Windows a **Primary Refresh Token (PRT)**.

1. Confirm SSO on the device: `dsregcmd /status` → **AzureAdJoined : YES** and **AzureAdPrt : YES**.

The user's Windows sign-in then flows silently into Edge/Office/Teams with no extra password.

Step 9b. Personal device — Microsoft Authenticator broker

1. On the personal device, the user installs **Microsoft Authenticator** (iOS/Android) — it is the **identity broker** that App Protection onboarding registers.

1. The first sign-in to a managed app (Outlook/Edge/Teams) registers the account through the broker; afterward, **other approved apps get silent SSO** with the same account (no re-login per app).

1. Do **not** require full device enrollment for SSO — the broker provides SSO while the device stays BYOD (App Protection only).

Step 9c. Confirm the same identity owns both

1. Entra → **Devices** → filter by the pilot user: confirm the **same UPN** has the corporate device (Entra joined) **and** a registered personal device under one identity.

1. The user should now move between their two devices and into M365 apps **without re-entering credentials** beyond the one sign-in per device.

Phase 6 — Pilot, validate, expand

Step 10. Pilot with one user on two devices

1. As a pilot user, enroll a **corporate** device (it tags Corporate → gets compliance/config).

2. On a **personal** device, sign into **Outlook/Edge** with the **same account** → the App Protection policy applies (PIN prompt, managed-app data containment).

1. Confirm the **same UPN** now owns both a Corporate and a Personal device in Intune.

Step 11. Validate

1. Run `DUALDEV-Discovery.ps1` (tenant readiness) and `DUALDEV-Validation.ps1 -User <upn>`:

confirm the user has **one Corporate** device (compliant) and **one Personal** device (app protection applied), under a single identity.

1. Confirm the **Corporate** filter applied device policy and the **Personal** filter did not (and vice-versa for app protection).

1. **Verify single sign-on:** on the corporate device `dsregcmd /status` shows **AzureAdPrt : YES**; on the personal device, after the first managed-app sign-in, a second approved app (e.g. Teams) opens **without** a fresh password prompt — silent SSO on both devices for the one account.

1. Review Conditional Access **report-only** results for the pilot user.

Step 12. Expand

1. When the pilot passes, widen the assignments from the pilot group to broader user groups.
2. Move **CA – COD-BYOD Access** from **report-only** to **On**.
3. Roll out in waves, validating each wave with the scripts before widening.

5. Testing & Validation — Dual-Device (Personal + Corporate) Management with Single Sign-On

Validate against a **pilot user with two devices** (one corporate, one personal) before expanding. Each test has an explicit **pass/fail gate**. Use the included PowerShell scripts (Microsoft Graph) for tenant-side checks and the on-device steps for the user experience.

Test 1 — Tenant readiness

Procedure: Run `DUALDEV-Discovery.ps1`. **Pass:** Enrollment restrictions allow the intended Personal/Corporate split, an App Protection policy exists, a compliance policy exists, Corporate/Personal **device filters** exist, Company Portal is assigned, and a Conditional Access policy is present. **Fail:** Any missing → resolve before piloting.

Test 2 — Single identity, two devices

Procedure: Run `DUALDEV-Validation.ps1 -User <upn>` for the pilot user. **Pass:** The **same UPN** owns at least one **Corporate** device and one **Personal** device. **Fail:** Devices split across different accounts → consolidate identity (Requirements R1).

Test 3 — Corporate device gets full MDM

Procedure: Inspect the pilot **corporate** device in Intune. **Pass:** Ownership = **Corporate**; the **compliance** and **configuration** policies (Corporate filter) are **applied**; device reports **Compliant**. **Fail:** Policies not applied → confirm the device is tagged Corporate (R5) and the Corporate filter is set on the assignment (Implementation §5–6).

Test 4 — Personal device gets App Protection only

Procedure: On the pilot **personal** device, sign into Outlook/Edge with the pilot account. **Pass:** The App Protection policy applies — **PIN prompt**, corporate data confined to managed apps, **Save As**/paste to personal locations blocked. The device is **not** fully enrolled (or, if personally enrolled, is tagged **Personal** and does **not** receive Corporate device policies). **Fail:** No PIN/containment → confirm the app is approved/managed and the App Protection policy is assigned to the user (Implementation §7).

Test 5 — Ownership filters route correctly

Procedure: Compare policy application across the two devices. **Pass:** The **Corporate** device receives device compliance/config and **not** the BYOD-only items; the **Personal** device receives app protection and **not** corporate device policies. **Fail:** Cross-application → re-check the `deviceOwnership` filter rules and

the Include/Exclude on each assignment (Implementation §4).

Test 6 — One account reaches work from both devices

Procedure: With Conditional Access **report-only**, access a corporate web app from the corporate device and a managed app from the personal device, signed in as the same user. **Pass:** Both succeed — the corporate device satisfies **compliant device**; the personal device satisfies **app protection policy** — under the **single** CA policy. **Fail:** One path blocked → review CA grant controls ("require one of"), compliance state, and app protection assignment (Implementation §9).

Test 7 — Unmanaged path is blocked (data protection holds)

Procedure: On the personal device, try to open corporate mail in the **built-in** (unmanaged) mail app or paste corporate data into a personal app. **Pass:** Corporate data is **not** accessible in unmanaged apps; paste/Save-As to personal locations is blocked. **Fail:** Corporate data leaks to unmanaged apps → tighten App Protection data-relocation settings.

Test 8 — Compliance & enforcement

Procedure: Review compliance for the corporate device and the report-only CA results for the user. **Pass:** Corporate device **Compliant**; CA evaluates the user correctly from both devices. **Fail:** Non-compliant or CA misfires → review the compliance policy and CA conditions.

Acceptance gate

Proceed to **Expand** only when Tests 1–8 pass for a pilot user who has **both** a corporate and a personal device signed in with the **same account**, and the data-protection checks (Test 7) hold.

6. Troubleshooting — Dual-Device (Personal + Corporate) Management with Single Sign-On

Work top-down: confirm the **identity** (one account), then **ownership** (Corporate vs Personal), then **targeting** (filters/assignments), then **access** (Conditional Access). Most issues are a device tagged with the wrong ownership, a missing filter on an assignment, or a BYOD user not using a managed app.

A user's two devices show under different accounts

- **Duplicate identities.** Consolidate to one Entra account; the user must sign in with the same UPN on both devices (Requirements R1). Re-enroll/re-register the stray device under the correct account.

A corporate device is tagged Personal (or vice-versa)

- **Wrong ownership.** Intune → **Devices** → **(device)** → **Properties** → **Device ownership** → set **Corporate**/**Personal**. For COD, add **corporate device identifiers** or use a corporate enrollment program so it tags Corporate automatically (R5).

- After fixing, the device picks up the correct filtered policies on next sync.

Corporate policies are not applying to the corporate device

- **Filter mismatch.** The compliance/config assignment must **Include** the **Corporate Devices** filter and target the **user** group (Implementation §5–6). Verify the filter rule is `(device.deviceOwnership -eq "Corporate")`.
- **Device not compliant/enrolled.** Confirm the device is enrolled and checking in.

App Protection isn't applying to the BYOD device

- **User not in scope.** App Protection assigns to the **user** group — confirm the pilot user is a member (Implementation §7).
- **App not managed/approved.** The user must use an **approved managed app** (Outlook, Edge, Teams).

Built-in mail/browser apps are intentionally excluded.

- **No Intune/MAM license** on the user (R2).
- App Protection applies on **next app launch / token refresh** — close and reopen the managed app.

BYOD user can't access corporate mail/data

- **By design in an unmanaged app.** They must use the managed app (e.g. Outlook), not the built-in one. Deploy and direct them to the approved apps (Implementation §8).
- **Conditional Access blocking.** If CA requires app protection, accessing from a non-managed app is blocked — expected. Check Entra → **Sign-in logs** for the CA result.

One account works on the corporate device but not the personal one (or vice-versa)

- **CA grant too strict.** The Conditional Access policy must **require *one of*** compliant device **or** app protection — not **both** (Implementation §9). "Require all" blocks one of the two paths.
- **App protection not yet registered** on the personal device — open a managed app and complete the PIN setup.

Personal device received corporate device policies (over-management)

- The personal device was **fully enrolled** and/or tagged **Corporate**. Confirm enrollment restrictions and ownership (R4–R5); for BYOD, prefer **App Protection without enrollment**. If personal MDM enrollment is intended, ensure it's tagged **Personal** so the Personal filter excludes corporate policies.

Conditional Access has unexpected effects

- Keep the pilot policy in **report-only** and read Entra → **Sign-in logs** → **Conditional Access** to see exactly which control was/wasn't satisfied before turning it **On**.

Useful checkpoints

- **Intune:** Devices → (device) → **Properties** (ownership), **Compliance**, **Configuration**; Apps → **App protection policies** → **(policy)** → **Monitor** for per-user status; Tenant administration → **Filters**.
- **Entra: Sign-in logs** → **Conditional Access** tab for grant evaluation.
- **Scripts:** re-run `DUALDEV-Validation.ps1 -User <upn>` after each change to confirm the gate.

7. Best Practices — Dual-Device (Personal + Corporate) Management with Single Sign-On

Field-proven guidance to keep one identity working cleanly across both device types.

Identity & assignment model

- **Assign policies to user groups, scope with device filters.** This is the heart of the design: one assignment to the user, the right policy follows the device's ownership. Avoid mixing per-device group assignments that fight the filters.
- **Keep one Entra identity per user.** Never create a separate "BYOD account" — it splits policy and breaks Conditional Access logic.
- **Name groups and filters by intent** (e.g. *PILOT – COD-BYOD Users*, *Corporate Devices*, *Personal Devices*) so assignments are self-documenting.

Ownership hygiene

- **Make ownership deterministic.** Use corporate enrollment programs and corporate device identifiers so COD tags **Corporate** automatically; let BYOD default to **Personal**.
- **Audit ownership regularly** with `DUALDEV-DeviceReport.ps1` — a device tagged the wrong way gets the wrong policy set.

BYOD data protection

- **Prefer App Protection (MAM) without enrollment for BYOD.** It protects corporate data without managing the employee's personal device — better adoption, less privacy friction, strong data-leak controls.

• **Lock the data boundary:** block backup, restrict cut/copy/paste and "Save As" to managed apps, require a PIN, and enforce encryption within managed apps.

• **Only enroll a personal device (MDM) when you truly need device settings** — and keep it tagged **Personal** so it never receives corporate device policies.

Corporate device management

• **Apply a real compliance baseline** (OS minimum, encryption, firewall) and gate access on it via Conditional Access.

• **Layer the zero-touch enrollment solutions** (Autopilot / Apple ADE / Android Enterprise) for hands-off COD provisioning; this solution unifies the model they all feed into.

Conditional Access

• **One policy, "require one of."** Grant access when the request is from a **compliant** device **or** an **app-protection-protected** app — never require both, or you lock out one device type.

• **Always start report-only.** Read the sign-in logs for the pilot user before enforcing.

• **Add MFA** as part of the same access story for a complete Zero Trust posture.

Rollout discipline

• **Pilot one user with two real devices** (a corporate and a personal one) — it's the only way to prove the single-identity experience end to end.

• **Expand in waves**, validating each with `DUALDEV-Validation.ps1` before widening.

• **Communicate the BYOD deal to employees:** what IT can and cannot see/do on a personal device under App Protection (data containment, *not* device surveillance) — clarity drives adoption.

Operations & documentation

• **Keep a runbook** of the pilot/production groups, the two ownership filters, the compliance and app-protection policies, and the Conditional Access policy — the moving parts that define the model.

• **Report monthly** with `DUALDEV-DeviceReport.ps1`: device counts by ownership, users who have both a COD and a BYOD device, compliance, and app-protection coverage.

8. Deliverables Summary — Dual-Device (Personal + Corporate) Management with Single Sign-On

Everything included in this solution package, and what each piece is for.

Documentation (this guide)

- **Executive Summary** — the business case and outcome in one page.
- **Solution Overview** — the two ownership models, the single-identity design, management models, benefits, and honest limitations.
- **Requirements Guide** — every prerequisite (one identity, MDM+MAM licensing, Entra ID P1, enrollment restrictions, corporate tagging, Company Portal, App Protection, compliance, ownership filters, network) with how to **verify** and how to **implement if missing**, plus the category-level Global Requirements.
- **Implementation Guide** — exact, ordered, click-by-click steps across six phases: identity & enrollment baseline, ownership-based targeting, COD full-MDM, BYOD App Protection, unifying access with Conditional Access, then pilot/validate/expand.
- **Testing & Validation** — eight tests with explicit pass/fail gates and an acceptance gate, proving one account works on both a corporate and a personal device.
- **Troubleshooting** — top-down diagnosis for identity, ownership, targeting, and access failures.
- **Best Practices** — the assign-to-user/scope-by-filter model, ownership hygiene, BYOD data protection, COD management, Conditional Access, and rollout discipline.

PowerShell scripts (Microsoft Graph) — `DUALDEV-Scripts.zip`

All scripts are tenant-side (run from an admin workstation with the Microsoft Graph PowerShell SDK); each has a header explaining **Purpose · How to run · Expected results**.

1. `DUALDEV-Discovery.ps1` — tenant readiness: enrollment restrictions (personal allowed?), compliance policies, App Protection policies, Corporate/Personal device filters, Company Portal, and Conditional Access presence.
1. `DUALDEV-Validation.ps1` — for a given user, confirms a single identity owns both a **Corporate** device (compliant) and a **Personal** device (app protection applied), with a PASS/FAIL summary.
1. `DUALDEV-DeviceReport.ps1` — tenant-wide CSV: devices by **ownership**, platform, enrollment type, compliance, and primary user — plus a list of users who have **both** a COD and a BYOD device.
1. `DUALDEV-Remediation.ps1` — safe remediation: set device **ownership**, re-assign policies to the target group, and flag enrollment-restriction/filter misconfigurations. Read-only by default; changes require an explicit confirmation switch.
1. `Set-OwnershipBasedTargeting.ps1` — creates the **Corporate** and **Personal** `deviceOwnership` filters (and optionally assigns the compliance/app-protection policies with those filters) — the mechanism that makes one account work cleanly across both device types.

Hero image

- `DUALDEV-hero.png` — the branded product image (1200×1200) used on the marketplace listing.

How it's delivered

A branded **Word** document and **PDF** (cover, table of contents, page numbers, and a scripts appendix), plus all five PowerShell scripts in a single **ZIP** — yours to keep and reuse across **unlimited devices**.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

DUALDEV-DeviceReport.ps1

Purpose: Exports every managed device with ownership (Corporate/Personal), platform, enrollment type, compliance, and primary user — then lists users who own BOTH a corporate and a personal device (the population this solution is designed for).

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\DUALDEV-DeviceReport.ps1 .\DUALDEV-DeviceReport.ps1 -CsvPath C:\Temp\cod-byod.csv

DUALDEV-Discovery.ps1

Purpose: Confirms the building blocks of the one-identity / two-ownership model exist: enrollment restrictions, a compliance policy (for COD), App Protection policies (for BYOD), the Corporate/Personal device filters, Company Portal, and a Conditional Access policy.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\DUALDEV-Discovery.ps1 Reads only — makes no changes.

DUALDEV-Remediation.ps1

Purpose: The common fixes that unblock the COD/BYOD model: correct a device's ownership (Corporate/Personal) so the right filtered policies apply, and audit for devices whose ownership is Unknown. Read-only by default.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once # Audit only (no changes): .\DUALDEV-Remediation.ps1 # Set a device's ownership (explicit opt-in): .\DUALDEV-Remediation.ps1 -Apply -DeviceName "Jane-Surface" -Ownership company

DUALDEV-Validation.ps1

Purpose: For a given user (UPN), lists their Intune devices and validates the acceptance gate: the same account owns at least one Corporate device (compliant) and at least one Personal device, and an App Protection policy targets the user. Produces a clear PASS/FAIL verdict.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\DUALDEV-Validation.ps1 -User jane@contoso.com

Set-OwnershipBasedTargeting.ps1

Purpose: Creates the two Intune assignment filters on `deviceOwnership` that make "one account, both device types" work: a Corporate filter (routes full-MDM policy to company devices) and a Personal filter (keeps corporate device policy off BYOD). Idempotent — skips a filter that already exists.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\Set-OwnershipBasedTargeting.ps1 -WhatIf # preview .\Set-OwnershipBasedTargeting.ps1 # create